

# Markov Chains (AI HL)



## Constructing a transition matrix

- 1 A weather model has two states, Sunny (S) and Rainy (R). From Sunny: 80% stay Sunny, 20% become Rainy. From Rainy: 40% become Sunny, 60% stay Rainy. Write down the transition matrix  $T$ , using columns for the 'from' state in the order (S, R).

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## Applying the transition matrix once

- 2 Today is sunny, so  $\mathbf{p}_0 = (1, 0)^T$  (order S, R). Using the matrix  $T$  from the previous question, find  $\mathbf{p}_1 = T\mathbf{p}_0$ , the probability distribution for tomorrow.

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## Applying the transition matrix twice

- 3 Using  $\mathbf{p}_1 = (0.8, 0.2)^T$  from the previous question, find  $\mathbf{p}_2 = T\mathbf{p}_1$ .

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## Finding the steady-state vector

- 4 Find the steady-state vector  $\mathbf{s} = (s_1, s_2)^T$  for the transition matrix  $T$  above, satisfying  $T\mathbf{s} = \mathbf{s}$  and  $s_1 + s_2 = 1$ .

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## Explaining why a Markov chain is regular

- 5 Explain why the Markov chain above converges to a unique steady state regardless of the initial distribution.

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## Computing and interpreting $T^2$

- 6 Using the transition matrix  $T$  with rows (0.8, 0.4) and (0.2, 0.6), find  $T^2$  and interpret its (1,1) entry.

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## Constructing a 3-state transition matrix

- 7 A game has three states: Win (W), Draw (D), Loss (L). From W: 50% stay W, 30% go to D, 20% go to L. From D: 40% go to W, 20% stay D, 40% go to L. From L: 10% go to W, 30% go to D, 60% stay L. Write down the transition matrix  $T$ , using columns for the 'from' state in the order (W, D, L).
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### Applying the transition matrix to a 3-state chain

- 8 Today's state is Draw, so  $\mathbf{p}_0 = (0, 1, 0)^T$  (order W, D, L). Using the matrix  $T$  above, find  $\mathbf{p}_1 = T\mathbf{p}_0$ .
- 9 Using  $\mathbf{p}_1 = (0.4, 0.2, 0.4)^T$  from the previous question, find  $\mathbf{p}_2 = T\mathbf{p}_1$ .
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### Defining the steady-state equation

- 10 State the equation that defines a steady-state vector  $\mathbf{s}$  of a transition matrix  $T$ , and explain what it means for  $\mathbf{s}$  to be an eigenvector of  $T$  with eigenvalue 1.
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### Verifying a steady-state vector directly

- 11 Verify that  $\mathbf{s} = (2/3, 1/3)^T$  is the steady-state vector for the weather transition matrix  $T$  above, by checking  $T\mathbf{s} = \mathbf{s}$  directly.
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### The memoryless (Markov) property

- 12 Explain what is meant by saying a Markov chain has the 'memoryless' property.